University of Glasgow | School of Computing Science

CSC3104: Cloud and Distributed Computing

Project: 2026-2027

Date issued	*Friday, 11 September 2026*
Proposal submission	*23:59 (11:59pm) Friday, 25 September 2026*
Progress Presentation	*13:00 – 16:00 (Tutorial), Tuesday, 29 October 2026*
Final submission deadline	*23:59 (11:59pm), Sunday, 15 November 2026*
Dell Cloud Native Award Presentation	*XX November 2026 (TBC with DELL)*

This is an assessment component for the Cloud and Distributed Computing module worth 30%.

Description

You are asked to *design* and *implement* a (smart) (intelligent) (reliable) (secure) cloud-based distributed system. You will then write a report in the form of a research *article* documenting your design and implementation. The implemented system should demonstrate some *original contribution* that you will need to argue in your report by comparing and contrasting it to related research and/or industrial work.

You are free to choose what will be the focus of your contribution. It can be a distributed (Gen)(AI) application utilising cloud computing and cloud native that introduce some novel functionality, for example, a distributed measurement system to give a network-wide view of performance; an integrated distributed system employing technologies ranging from cloud computing, artificial intelligence and embedded system; a distributed data collection and analytic framework for environmental monitoring.

Your project **MUST** incorporate the **cloud native framework** such as **container and/or Kubernetes**, applying what you have learnt from the Dell's online courses in this module.

You will undertake this assignment *in groups* to accommodate for the extra load of research in deriving your own specification and design for the system you will be implementing. It is the group members' responsibility to divide the work and manage the overall project.

The following list contains a few *past examples* of what can constitute an original contribution but it is by no means exhaustive:

- A smart distributed sensor-actuator application that offers intelligent automation in a home area network.
- Implementation of a (new) consensus algorithm for a mini-Blockchain
- Building a Raspberry Pi cloud (https://ieeexplore.ieee.org/document/6679872) (https://raspberrypicloud.wordpress.com/)
- Porting an existing cloud-based distributed system to Google RPC (https://grpc.io)
- Adding distribution and containerise some centralised/monolithic systems, implement microservice architecture

- Distributed indoor localisation system for robots.
- Implementation of federated learning with an emphasis on distributed reliability and privacy preservation
- Adding redundancy to an existing distributed computation to ensure e.g. correctness through voting or better performance
- An implementation of a novel distributed algorithm for communication or coordination (e.g. leader election, mutual exclusion, multicast, etc.)
- An implementation of a distributed algorithm with the help from cloud computing that tackles an unsolved problem/works under certain conditions where other (equivalent) algorithms fail (e.g. by converting certain types of failures to more acceptable types or avoiding them altogether)

You will have to argue about the originality of your contribution in your article.

The following events and outlets can give you an idea of research contributions in cloud & distributed systems and related areas. Some of the systems/approaches described therein may seem overly complex (your coursework can be something much simpler!) but they should still give you good inspiration for the kinds of systems you can focus on as well as an idea of how to shape your article. You should definitely do some research at these places (and others you can find online) before initiating discussions within your group about the system/functionality you will be developing.

ACM/IFIP/USENIX International Middleware Conferences:

https://middleware-conf.github.io/2024/program/full-program/

IEEE International Parallel & Distributed Processing Symposium

https://www.ipdps.org/

IEEE International Conference on Cloud Computing

https://cloud.conferences.computer.org/2024/
https://conferences.computer.org/cloud/2023/

IEEE Pervasive Computing conferences:

https://www.percom.org/ or https://ieeexplore.ieee.org/xpl/conhome/1000551/all-proceedings

https://www.sigmobile.org/mobisys/

IEEE World Forum on Internet of Things (WF-IoT)

https://wfiot2024.iot.ieee.org/or https://wfiot2023.iot.ieee.org/

IEEE Transactions on Parallel and Distributed Systems (TPDS):

https://ieeexplore.ieee.org/xpl/RecentIssue.jsp?punumber=71

Elsevier Journal of Parallel and Distributed Computing:

https://www.elsevier.com/wps/find/journaldescription.cws_home/622895/description

You should be able to access papers from these outlets via the Glasgow University's electronic subscriptions (e.g. to IEEEXplore and the ACM Digital Library) using your GUID, or SIT University Network (access through SIT VPN).

Each group is required to submit a project proposal (no more than 2 pages) in Week 4 to outline your initial idea, motivation, problems and challenges that the proposal is trying address. The proposal is important to allow for the lecturer to provide early feedback on the idea, so that it can be improved further. Individual group consultation sessions can be organised as part of the tutorial/lab sessions to discuss the project ideas.

Your code can be written in any programming languages, such as Java, Python, C++ and **it must use Container and/or Kubernetes**. Marks will be awarded based on the elegance of your implementation, the level of challenge involved in realising it, and features included such as e.g. handling concurrency (interference, memory consistency etc.), use of cloud services, concurrent tasks for e.g. housekeeping. You must containerise (part of) your project and test your code before submission; excuses such as "it worked at home" are not acceptable.

Your final article should be around 3,000 words and no more than 6 pages long following either the IEEE or the ACM proceedings format. MS Word and LaTeX templates can be found at the following locations (feel free to increase font to 11pt.), respectively:

http://www.ieee.org/conferences_events/conferences/publishing/templates.html

https://www.acm.org/publications/proceedings-template

The final article should contain at least the following sections (title, additional sections, sub-sections, numbering and actual heading names are left at your discretion):

Abstract: A summary of your contribution; what problem does your system solve; what new functionality it enables; what you have achieved by building it; no more than 200 words.

Introduction (and motivation): Put your work into a wider context; give a brief history of the area and the state of the art; elaborate on your choice of building the particular system; justify why this system is important and why it is needed (expand on the main points sententiously covered in the abstract and explain what the work documented in the article is all about).

System Design: Describe your design choices and give a high-level overview of your system and its functionality; include figures and/or diagrams to explain concepts, interactions between components etc.; justify your design choices and discuss any resulting implications

Implementation: Describe the details of your implementation; how you have realised your design; what aspects of it worked as expected and whether there has been any missing functionality; how you have dealt with important issues such as concurrency, server-to-client communication, fault and failure handling, etc.

Evaluation: Provide evidence that your system works; include quantitative indicators that verify any claims you have made; provide proof of the performance of your system; use graphs to visualise your main results.

Related Work: *Critically* describe related work in the field you chose to focus on and highlight the most important contributions made by others; compare and contrast existing work with the system you have implemented and therefore highlight your main achievements; say what existing systems do not do that your system does.

Conclusions (and future work): Provide a brief description of the overall system you have documented and highlight your main contributions. Include a main take-away message and possibly identify future dimensions for your work.

References: Provide a numbered list of related works you have identified and discussed; it should include articles and possibly on-line resources; this is not a bibliography, only include works that you have cited from within your article.

Assessment

Proposal (17%): Marks will be awarded for coherent proposal that clearly articulates the application domain(s) of the project, problems and challenges and how cloud and distributed algorithms can be used to address them. The proposal must clearly describe the current *state-of-the-art* (i.e., what are the existing approaches in designing and implementing the proposed application). Finally, a sketch of your idea, outlining the approach and algorithms that you plan to implement for the proposed project.

Progress Presentation (15%): This is a mid-point check on your project's progress. Marks will be allocated based on the progress made. You will be given 5-10 mins of presentation time in the tutorial session and each group member is expected to explain his/her contribution.

Implementation (25%): Marks will be awarded for working systems that implement the advertised functionality. Level of marks will depend on the originality and contribution of the system and the level of challenge involved in realising it. Inclusion of advanced features such as concurrency control, serverless architecture, server callbacks, etc. will also be taken into consideration. You are NOT expected to provide any sophisticated (G)UI for your system. Your solution should mainly focus on aspects of the system that make it functions in a cloud native and distributed fashion.

Presentation (18%): Each group will be required to submit a short video of 10 to 15-minute in duration, describing their implementation, detailing the functionalities and the design justification of the distributed application they have developed. ****The top 3 project teams will be invited to present their project to a panel of judges to be considered for the Dell Cloud Native Award**.

Article/Report (25%): Marks will be awarded for coherent articles that will satisfactorily address the main points of the relevant sections outlined above. Particular attention should be given to the construction of the argument regarding the originality of the implemented system and its comparison to related work, as well as to the quantitative evaluation of the system.

***Note: Each member be required to submit a peer evaluation score to rate the other members' contribution at the end of the project. The final mark will be adjusted based on the peer evaluation score.**

What to submit

The proposal (max. 2 pages) – a single .pdf file called GROUPNN.pdf, will be submitted *electronically* via LMS **no later than 23:59 (11:59pm), Friday, 25th September 2026.** A submission link will be provided on LMS Cloud and Distributed Computing page. Only one submission per group.

The final mini project will be submitted *electronically* via LMS **no later than 23:59pm (11:59pm), Sunday, 15th November 2026.** A submission link will be provided on LMS Cloud and Distributed Computing page for the following items:

- Softcopy version of your system – all of the *source code* for your system (.java .cpp .py, launch scripts, test data, etc.) should be packaged into a Docker container image to be submitted or specify an URL or github repo.
- Softcopy version of your *article* – a single .pdf file called GROUPNN.pdf where NN is your group number.
- Softcopy version of your *testing instructions* – a single .pdf file containing detailed instructions of how you expect me to test your system. You should be explicit about steps I should take for testing your program. The file should be called GROUPNN_INSTRUCTIONS.pdf, where NN is your group number.
- Video presentation (you can either upload your video onto xSite or specify a URL/youtube link to access your video).

One submission for each of the aforementioned *four files* should be made per group by one of the members.